Deep Reinforcement Learning

Value function approximation

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Content

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- Limitations of tabular methods
- Value estimation / prediction with function approximation
- Gradient-based prediction
- Batch learning

Where are we?

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	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6	Brief introduction to deep learning	The basics for what comes next
7	Value function approximation	Value estimation with function approximation
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Lecture contents

Limitations of tabular methods

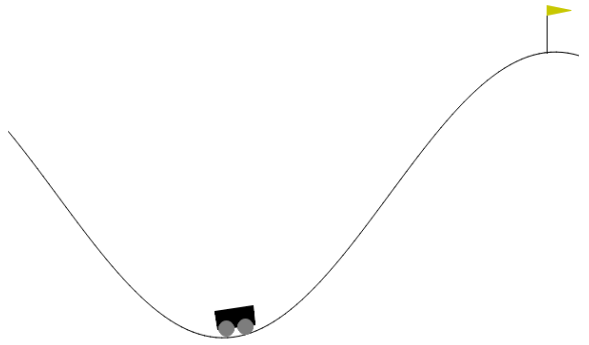
Limitations of tabular methods

We have essentially "solved" the RL problem in part one of our lecture! But there are quite a few limitations when it comes to real-world systems:

- The **curse of dimensionality**: For large dimensions $|\mathcal{S}|$ or $|\mathcal{A}|$ (even moderate ones), the calculations become prohibitively expensive and convergence is very slow.
- **Continuous state spaces**, i.e., infinitely many possible states

$$s \in \mathcal{S} \subseteq \mathbb{R}^n \quad / \quad |\mathcal{S}| = \infty. \qquad (\text{instead of } \mathcal{S} = \{s_1, \ldots, s_{|\mathcal{S}|}\})$$

  - **Example**: The *mountain car*, with continous $\mathcal{S} = [x_{\text{min}}, x_{\text{max}}] \times [v_{\text{min}}, v_{\text{max}}]$ (minimal and maximal position x and velocity v), but finite $\mathcal{A} = \{-1, 0, 1\}$ (accelerate left, do nothing, accelerate right).

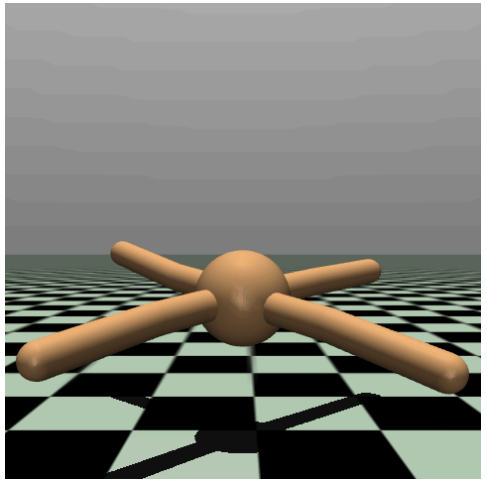


Mountain car [Source].

- **Continuous action spaces**, i.e., infinitely many possible actions*

$$a \in \mathcal{A} \subseteq \mathbb{R}^m \quad / \quad |\mathcal{A}| = \infty. \qquad (\text{instead of } \mathcal{A} = \{a_1, \ldots, a_{|\mathcal{A}|}\})$$

  - **Example**: The *ant*, with 27 continous state variables and 8 continuous action variables.



Ant [Source].

* For now, however, we will stick to finite action spaces.

Value estimation / prediction with function approximation

Value estimation / prediction with function approximation

- From now on, we will introduce function approximators for the quantities of interest.
- For the value function, this means that we're looking for

$$V_\theta(s) \approx V^\pi(s).$$

  - $V_\theta(s)$ may be a *linear model*, i.e.,

$$V_\theta(s) = \sum_{k=1}^{d} \theta_k \psi_k(s) = \theta_1 \psi_1(s) + \ldots + \theta_d \psi_d(s),$$

  - but also a *nonlinear model* such as a deep neural network.
- The weight vector $\theta \in \mathbb{R}^d$ typically is of lower dimension d than the state space $\mathcal{S}$: $d \ll |\mathcal{S}|$.
  - Otherwise this exercise would be pointless!
  - This obviously holds for continuous state spaces, but also for very large yet finite-dimensional state spaces.

A note on the state space dimension

We use n to denote the number of state variables, but this does not mean that we need to have $n = \infty$ for continuous state spaces. A state space is finite if each state variable s_k is an element of a finite set, whereas it is continous, if at least one $s_1, \ldots, s_n$ is a continuous variable.

Global impact of experience

Updates due to new experience lead to updates in our model parameter θ. This means that *we modify our value estimate for many states*, in contrast to tabular methods where we update individual states only.

💡 We will use greek letters (e.g., θ or ϕ) to indicate the trainable parameters.

Prediction objective

- In the tabular case a specific prediction objective was not needed:
  - The learned value function could exactly match the true value.
  - The value estimate at each state was decoupled from other states.
- Due to global impact of experience, we need to define an accuracy metric on the entire state space: the **RL prediction goal**.

Definition: Value error (VE)

The RL prediction objective is defined as the mean squared value error

$$\overline{VE}(\theta) = \int_{\mathcal{S}} \mu(s) \Big(V^{\pi}(s) - V_{\theta}(s) \Big)^2 \, \mathrm{d}s.$$

Here, $\mu(s) \geq 0$ is the *state distribution*, with $\int_{\mathcal{S}} \mu(s) \, \mathrm{d}s = 1$.

Challenge:

- The true value $V^{\pi}(s)$ is unknown in most tasks!
- We need to find ways to estimate $V^{\pi}(s)$ from experience.

The on-policy prediction objective

- For prediction we will **focus entirely on the on-policy case**.
- Hence, $\mu(s)$ is the on-policy distribution under the current policy π.

On policy prediction objective

As $\overline{VE}(\theta)$ is the expected error, we can use *Monte Carlo sampling* to estimate it:

$$L(\theta) = \sum_{k=1}^{N} \Big(V^{\pi}(s_k) - V_{\theta}(s_k) \Big)^2 \approx \overline{VE}(\theta). \tag{1}$$

Goal: find the optimal value estimate V_{θ^*} by optimization:

$$\theta^* = \arg\min_{\theta} L(\theta).$$

Challenges:

1. The true value $V^{\pi}(s)$ is unknown in most tasks!
2. The function approximator $V_{\theta}(s)$ needs to be able to approximate $V^{\pi}(s)$.
3. Training:

- If $V_{\theta}(s)$ is a linear model: convex optimization problem.
  - The “easy” case: local optimum = global optimum, uniquely defined.
- If $V_{\theta}(s)$ is nonlinear model: nonlinear optimization problem.
  - The “hard” case: many local optima with no guarantee to locate the global one.
  - Training may diverge, advanced optimization techniques are very important!

💡 **Off-policy prediction**: *importance sampling* to transform the sampled value estimates from the behavior to the target policy.
⇒ Increases the prediction variance; in combination with generalization errors due to function approximation: large risk of diverging.

Gradient-based prediction

Incremental weight updates

- Now that we have an objective $L(\theta)$ we would like to optimize for (Eq. (1)), let's use **gradient descent** with learning rate α:

$$\theta^{(t+1)} = \theta^{(t)} - \frac{1}{2}\alpha\nabla L\left(\theta^{(t)}\right) = \theta^{(t)} - \frac{1}{2}\alpha\nabla_\theta\left(\sum_{k=1}^{N}\left(V^\pi(s_k) - V_\theta(s_k)\right)^2\right).$$

- What's the problem here?
  $\Rightarrow$ We often only have a *single sample*. (Or, instead, we would like to perform an update after a single sample.)

SGD-based parameter update

Given a new sample s_t and its (for now exact) value $V^\pi(s_t)$, we can update our current weight vector $\theta^{(t)}$ by using **stochastic gradient descent** (**SGD**) with:

$$\theta^{(t+1)} = \theta^{(t)} - \frac{1}{2}\alpha\nabla_\theta\left[\left(V^\pi(s_t) - V_\theta(s_t)\right)^2\right] = \theta^{(t)} + \alpha[V^\pi(s_t) - V_\theta(s_t)]\nabla_\theta V_\theta(s_t). \tag{2}$$

Question: Should we follow (2) to optimality?

$\Rightarrow$ No! We are optimizing for a single sample $\rightarrow$ **overfitting**.

$\Rightarrow$ Take a single, small step only, then collect new experience.

Bootstrapping the true value

Challenge 1. remains: We do not know $V^\pi(s_t)$ which we need for our prediction objective (1).

What can we do? $\Rightarrow$ Use an estimate for $V^\pi(s_t)$ that we can compute from experience!

Possible options for estimates:

Unbiased estimates of $V^\pi(s_t)$

Monte Carlo estimates for g_t, sample from entire trajectories.

Bootstrapped estimates of $V^\pi(s_t)$

- The Dynamic Programming target

$$V_\theta(s_t) \leftarrow \sum_{a \in \mathcal{A}} \pi(a|s_t) \sum_{s' \in \mathcal{S}} p\left(s'|s_t, a\right) \left[r + \gamma V_\theta(s')\right].$$

- TD targets (e.g., TD(0))

$$V_\theta(s_t) \leftarrow V_\theta(s_t) + \alpha \left[r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)\right].$$

💡 Inserting the bootstrapped versions into (2) does not yield a true gradient descent method, as we are ignoring the parametric dependency of the target on θ (Sutton and Barto 1998). More on this on the next slides.

Algorithm: Gradient Monte Carlo value estimation

Algorithm: Gradient Monte Carlo algorithm for estimating $V_\theta \approx V^\pi$

Input:

- The current policy π
- a differentiable, parameter-dependent function $V_\theta : \mathcal{S} \to \mathbb{R}$
- learning rate α

Initialize: Value function weights $\theta \in \mathbb{R}^d$ arbitrarily

for $k = 1, 2, \ldots, K$ episodes:

Generate a sequence following π:

$$((s_0, a_0, r_0), (s_1, a_1, r_1), \ldots, (s_{T_k-1}, a_{T_k-1}, r_{T_k-1}))$$

Calculate the every-visit returns g_t

for $t = 0, 1, \ldots, T-1$:

$\theta \leftarrow \theta + \alpha[g_t - V_\theta(s_t)]\nabla_\theta V_\theta(s_t)$

- Direct transfer from the tabular case to function approximation.
- Only change: The entry-based update ($V(s_t) = \textbf{average}(\ell(s_t))$) is replaced by the SGD update for θ.

Semi-gradient methods

- If bootstrapping is applied, the true target $V^\pi(s_t)$ is approximated by a target depending on the estimate $V_\theta(s_t)$.
- If $V_\theta(s_t)$ does not fit $V^\pi(s_t)$ (which it does not before convergence), then the update target becomes a *biased estimate*.
  - For example, in the TD(0) case, we get for (1):

$$L(\theta) = \sum_{t=1}^{T} \Big(r_t + V_\theta(s_{t+1}) - V_\theta(s_t) \Big)^2.$$

  - Taking the gradient of the single-sample version for $L(\theta)$ yields:

$$\begin{aligned} \theta \leftarrow \theta - \frac{1}{2}\alpha \nabla_\theta \Big[(r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t))^2 \Big] \\ = \theta - \alpha [r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \nabla_\theta [\gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \\ \neq \theta + \alpha [r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \qquad \nabla_\theta V_\theta(s_t) \qquad \text{(Eq. (2))}. \end{aligned}$$

  - Application of Eq. (2) is still very common.
  - These appraoches are known as **semi-gradients** because they do not approximate the actual gradient.

TD(0) semi-gradient update of θ

$$\theta \leftarrow \theta + \alpha [r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)] \nabla_\theta V_\theta(s_t). \tag{3}$$

💡 Even though widely applied, there are no convergence guarantees for semi-gradient methods (Sutton and Barto 1998).

Algorithm: Semi-gradient TD(0) value estimation

Algorithm: Semi-gradient TD(0) for estimating $V_\theta \approx V^\pi$

Input:

- The current policy π
- a differentiable, parameter-dependent function $V_\theta : \mathcal{S} \to \mathbb{R}$
- learning rate α

Initialize: Value function weights $\theta \in \mathbb{R}^d$ arbitrarily

for $k = 1, 2, \ldots, K$ episodes:
  Initialize s_0
  for $t = 0, 1, \ldots, T-1$:
    Apply action $a_t \sim \pi(\cdot|s_t)$
    Observe r_t and s_{t+1}
    $\theta \leftarrow \theta + \alpha[r_t + \gamma V_\theta(s_{t+1}) - V_\theta(s_t)]\nabla_\theta V_\theta(s_t)$
    if s_{t+1} is **terminal** **then** STOP

Note: n-step version TD(n)

The above algorithm can be extended towards n-step bootstrapping by replacing the TD(0) target $r_t + \gamma V_\theta(s_{t+1})$ with the n-step bootstrapped return

$$\begin{aligned} g_{t:t+n} =& r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \ldots \\ &+ \gamma^{n-1} r_{t+n-1} + \gamma^n V_\theta(s_{t+n}). \end{aligned}$$

The most challenging part in the implementation is the assessment of the remaining number of steps (i.e., when $T - t < n$).

For details, see (Sutton and Barto 1998, Ch. 9.4).

Batch learning

Batch learning / experience replay

- As discussed during the TD learning lecture, training on individual samples can be slow.
- This also applies to SGD-based updates with bootstrapping.
- Alternative: Batch learning methods

Mini-batch semi-gradient TD(0) with experience replay

Based on a dataset $\mathcal{D} = \{(s_0, V^\pi(s_0)), \ldots, (s_N, V^\pi(s_N))\}$, repeat

- Sample uniformly b state-value pairs from experience (so-called mini batches), i.e., $\mathcal{B} = \{(s_i, V^\pi(s_i))\}_{i=1}^{b} \subset \mathcal{D}$:

$$(s_i, V^\pi(s_i)) \sim \mathcal{D}.$$

- Apply (semi) SGD update step:

$$\theta \leftarrow \theta + \frac{\alpha}{b} \sum_{(s_i, V^\pi(s_i)) \in \mathcal{B}} [V^\pi(s_i) - V_\theta(s_i)] \nabla_\theta V_\theta(s_i).$$

Remarks:

- Universally applicable: V_θ can be any differentiable function.
- The true target V^π is usually approximated by MC or TD targets.

Least squares policy evaluation

Linear modeling of value functions

- Let's assume that we choose an approximation that is *linear* w.r.t. the weights $\boldsymbol{\theta}$:

$$V_{\boldsymbol{\theta}}(s) = \boldsymbol{\theta}^\top \boldsymbol{\psi}(s) = \sum_{k=1}^{d} \theta_k \psi_k(s).$$

- What's the gradient in this case? $\Rightarrow$ it's simply the feature vector,

$$\nabla_{\boldsymbol{\theta}} V_{\boldsymbol{\theta}}(s) = \nabla_{\boldsymbol{\theta}} \boldsymbol{\theta}^\top \boldsymbol{\psi}(s) = \boldsymbol{\psi}(s).$$

- Our semi-gradient TD($\mathbf{0}$) update (3) thus simply becomes

$$\begin{aligned}
\boldsymbol{\theta} \leftarrow & \boldsymbol{\theta} + \alpha[r_t + \gamma V_{\boldsymbol{\theta}}(s_{t+1}) - V_{\boldsymbol{\theta}}(s_t)] \nabla_{\boldsymbol{\theta}} V_{\boldsymbol{\theta}}(s_t) \\
&= \boldsymbol{\theta} + \alpha \big[r_t + \gamma \boldsymbol{\theta}^\top \boldsymbol{\psi}(s_{t+1}) - \boldsymbol{\theta}^\top \boldsymbol{\psi}(s_t)\big] \boldsymbol{\psi}(s_t) \\
&= \boldsymbol{\theta} + \alpha \big[r_t + \gamma \boldsymbol{\theta}^\top \boldsymbol{\psi}_{t+1} - \boldsymbol{\theta}^\top \boldsymbol{\psi}_t\big] \boldsymbol{\psi}_t = \boldsymbol{\theta} + \alpha \big[r_t + \boldsymbol{\theta}^\top (\gamma \boldsymbol{\psi}_{t+1} - \boldsymbol{\psi}_t)\big] \boldsymbol{\psi}_t,
\end{aligned}$$

  where we have introduced $\boldsymbol{\psi}_t$ for $\boldsymbol{\psi}(s_t)$.

- Similarly, in **batch mode**, our bootstrapped version of the on-policy prediction objective (1) becomes

$$L(\boldsymbol{\theta}) = \sum_{t=0}^{T-1} \Big(r_t + \gamma \boldsymbol{\theta}^\top \boldsymbol{\psi}(s_{t+1}) - \boldsymbol{\theta}^\top \boldsymbol{\psi}(s_t)\Big)^2 = \sum_{t=0}^{T} \Big(r_t - \big[\boldsymbol{\psi}_t^\top - \gamma \boldsymbol{\psi}_{t+1}^\top\big] \boldsymbol{\theta}\Big)^2.$$

Least Squares TD learning (LSTD)

- Collect data:

$$y = \begin{bmatrix} r_0 \\ \vdots \\ r_{T-1} \end{bmatrix}, \qquad \Psi = \begin{bmatrix} \psi_0^\top - \gamma \psi_1^\top \\ \vdots \\ \psi_{T-1}^\top - \gamma \psi_T^\top \end{bmatrix}.$$

- Loss function:

$$L(\theta) = \sum_{t=0}^{T-1} \left(r_t - \left[\psi_t^\top - \gamma \psi_{t+1}^\top \right] \theta \right)^2 = \frac{1}{N} \| y - \Psi \theta \|_F^2.$$

- Optimal solution (a.k.a. **TD fixed point** (Sutton and Barto 1998, Sec. 9.4)):

$$\theta^* = (\Psi^\top \Psi)^{-1} \Psi^\top y = \Psi^\dagger y.$$

- Estimated value function:

$$V_{\theta^*}(s) = {\theta^*}^\top \psi(s) = \sum_{k=1}^{d} \theta_k^* \psi_k(s).$$

Recall: Linear regression

- Model: $\hat{y} = f_\Theta(x) = \Theta^\top x$.
- Loss function $L(\Theta)$ (MSE):

$$\frac{1}{N} \sum_{k=1}^{N} \left\| \Theta^\top x_k - y_k \right\|_2^2 = \frac{1}{N} \| X\Theta - Y \|_F^2.$$

- Batch data:

$$X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_N^\top \end{pmatrix}, \quad Y = \begin{pmatrix} y_1^\top \\ \vdots \\ y_N^\top \end{pmatrix}$$

- Optimal solution (Θ^*):

$$\Theta^* = (X^\top X)^{-1} X^\top Y = X^\dagger Y.$$

Summary / what you have learned

Summary / what you have learned

- To cover large or continuous state spaces **function approximation** is required.
- **On-policy prediction** is straightforward with function approximation:
  - Transfer of incremental learning from the tabular case to **gradient descent for θ**.
  - **Stochastic gradient descent** allows step-by-step based updates.
- Gradient-based prediction is **not risk free** (especially non-linear case):
  - challenging optimization problems (descent directions, learning rates, etc.),
  - local optima vs. global optimum.
- If **bootstrapping** is applied, the update target depends on θ.
  - True gradient becomes computationally more complex.
  - **Semi-gradient** methods reduce computational burden at accuracy costs.
- **Batch learning** is often more efficient for training.
- **Linear models** can be trained in a single step using the pseudo-inverse.

References

Sutton, R. S., and A. G. Barto. 1998. *Reinforcement Learning: An Introduction*. MIT Press.